

# Smart Student Expense Tracker

## ICA-03 — Web Services and Technology (IT2234)

**Module:** Web Services and Technology | Level: 2nd Year IT

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**GitHub Repository:**

[IT2234-Project-Assignment-Smart-Student-Expense-Tracker](#)

## Technologies

- Node.js
- Express.js
- MongoDB
- React
- Postman

# Problem Identification

## The Problem

- Many students do not track daily spending
- Monthly budgets are often ignored
- No clear view of where money goes
- Risk of overspending, especially when living away from home

## Target Users

- University undergraduates
- Students managing their own finances

## Why It Matters

- Better financial awareness
- Encourages responsible spending
- Supports real student life challenges

# Proposed Solution & Features

## Proposed Solution

A MERN-based expense tracking system where students can:

- Register and log in securely using JWT
- Record income and expenses with categories
- Set and monitor monthly budgets
- View summaries and charts on a dashboard

| Area           | Features                                    |
|----------------|---------------------------------------------|
| Authentication | Register, Login, JWT Protected Routes       |
| Transactions   | Full CRUD Operations                        |
| Budgets        | Set Monthly Budget, Track Remaining Balance |
| Analytics      | Summary API, Category Breakdown, Recharts   |
| Testing        | Postman Collection for All Endpoints        |

# System Architecture & Database

## High-Level Architecture

React (Vite) → Axios → Express REST API → MongoDB  
UI HTTP JWT Auth Collections

## Backend Structure

routes → controllers → models → MongoDB

## Database Collections

Users, Transactions & Budgets

## Security

Password hashing using bcryptjs, JWT stored in Authorization header and Authorization: Bearer TOKEN

# REST API & CRUD Demo

## API Base URL

<http://localhost:5000>

## Key API Endpoints

| Method                    | Endpoint                          | Description         |
|---------------------------|-----------------------------------|---------------------|
| POST                      | /api/auth/register                | Register User       |
| POST                      | /api/auth/login                   | Login & Get JWT     |
| POST / GET / PUT / DELETE | /api/transactions                 | Transaction CRUD    |
| GET                       | /api/transactions/summary         | Totals & Categories |
| POST / GET / PUT          | /api/budgets                      | Budget Management   |
| GET                       | /api/budgets/status?month=YYYY-MM | Budget Status       |

# Conclusion & Evaluation Alignment

## What Was Achieved

- Solved a real-world student finance problem
- Built complete RESTful backend with MongoDB
- Implemented mandatory CRUD operations
- Added validation, JWT authentication, and error handling
- Created Postman tests and GitHub documentation
- Developed React frontend with charts and dashboard

## Challenges Faced

- Connecting frontend to protected APIs
- Designing MongoDB schemas using userId
- Testing and debugging API endpoints
- Managing frontend state and API calls

## Future Improvements

- Email verification
- Password reset
- Export reports as PDF/CSV
- Mobile-responsive PWA version

**Thank you!**